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Rock Mechanics (MIN E – 620)

Homework No. 1



1 Homework No. 1

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A borehole deformation gauge test is conducted within a drill hole initially 5.0 cm in diameter. Young's Modulus is known to be 6.897×10^4 MPa. The U1 axis of the gauge was installed parallel to the true horizontal direction. The following diametral displacement data were measured following overcoring:

U1 = 1120 micro cm
U2 = 4540 micro cm
U3 = 3340 micro cm

1.1 Q1. Calculate the principal stresses and their orientation

Based on the known information, we can get Mohr's circle as follows:

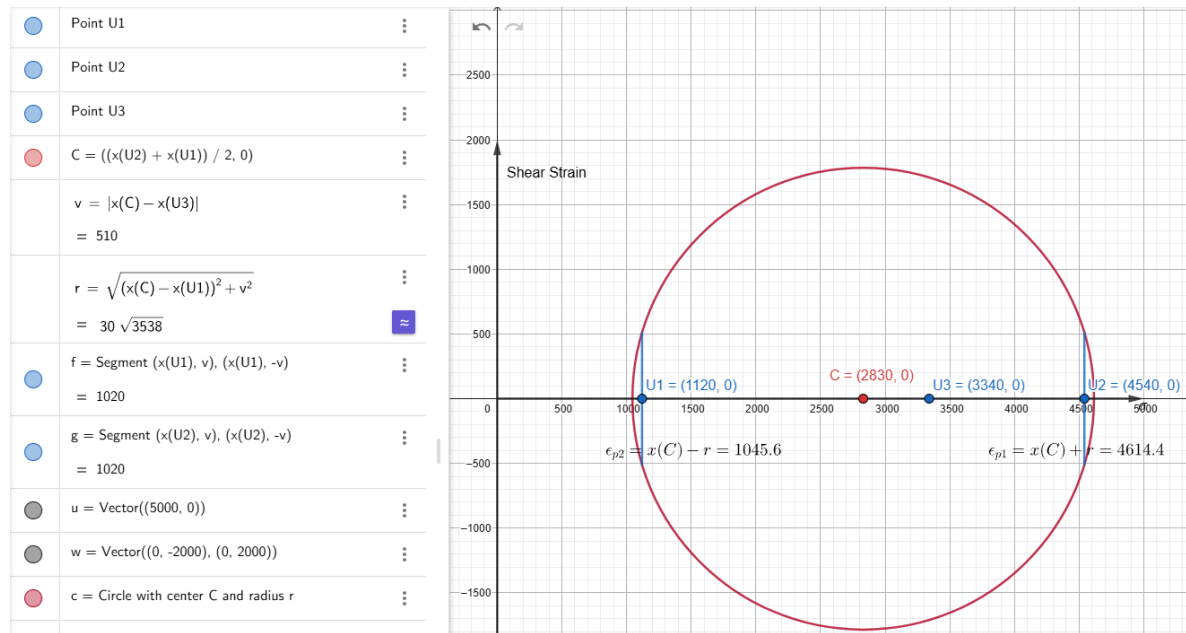


Figure 1: Mohr's Circle

Due to a lack of information about Poisson's ratio, we can only use the following equations (USBM Deformation Gage Test) to calculate the principal stresses:

$$\sigma_1 = \frac{E}{6d} \left[(U_1 + U_2 + U_3) + \frac{\sqrt{2}}{2} \sqrt{(U_1 - U_2)^2 + (U_2 - U_3)^2 + (U_3 - U_1)^2} \right] \quad (1)$$

$$\sigma_3 = \frac{E}{6d} \left[(U_1 + U_2 + U_3) - \frac{\sqrt{2}}{2} \sqrt{(U_1 - U_2)^2 + (U_2 - U_3)^2 + (U_3 - U_1)^2} \right] \quad (2)$$

$$\theta_p = 0.5 \tan^{-1} \left(\frac{\sqrt{3}(U_2 - U_3)}{2U_1 - U_2 - U_3} \right) \quad (3)$$

For convenient computation, set $S = U_1 + U_2 + U_3$, and $D = \frac{\sqrt{2}}{2} \sqrt{(U_1 - U_2)^2 + (U_2 - U_3)^2 + (U_3 - U_1)^2}$

$$S = 1120 + 4540 + 3340 = 9000 \text{ micro cm}$$

$$D = \frac{\sqrt{2}}{2} \sqrt{(1120 - 4540)^2 + (4540 - 3340)^2 + (3340 - 1120)^2} \approx 3005.4$$

Substitute $E = 6.897 \times 10^4$ MPa, and $d = 5$ cm into Equation 1 and Equation 2 and Equation 3 (1 micro cm = 10e-6 cm),

$$\begin{aligned} \sigma_1 &= \frac{6.897 \times 10^4}{6 \times 5} (9000 \times 10^{-6} + 3005.4 \times 10^{-6}) \\ &= 27.6 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \sigma_3 &= \frac{6.897 \times 10^4}{6 \times 5} (9000 \times 10^{-6} - 3005.4 \times 10^{-6}) \\ &= 13.78 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \theta'_p &= 0.5 \tan^{-1} \left(\frac{\sqrt{3}(4540 - 3340)}{2 \times 1120 - 4540 - 3340} \right) \\ &= -10.115^\circ \end{aligned}$$

For the various conditions, we have the rule as shown below,

For the various values of diametral deflection,
a range of (θ_p) angles can be determined

- $U_2 > U_3$ and $U_2 + U_3 < 2U_1$, then $(\theta_p) = 0-45\text{deg}$
- $U_2 > U_3$ and $U_2 + U_3 > 2U_1$, then $(\theta_p) = 45-90\text{deg}$
- $U_2 < U_3$ and $U_2 + U_3 > 2U_1$, then $(\theta_p) = 90-135\text{deg}$
- $U_2 < U_3$ and $U_2 + U_3 < 2U_1$, then $(\theta_p) = 135-180\text{deg}$

Figure 2: The rule for determining the angle range

Due to $U_2 > U_3$ and $U_2 + U_3 > 2U_1$, so

$$\theta_p = 90 - |\theta'_p| = 79.89^\circ$$

1.2 Q2 Does this test show the true orientation of the in-situ principal stresses, and if it doesn't, what would you have to do to measure the true 3-dimensional orientation of the principal stresses?

No, it doesn't.

This result couldn't show the true orientation of the in-situ principal stresses. Because the real in-situ stresses are affected by the 3D environment, this deformation gauge test only measured the plane perpendicular to the borehole, which is a projection of the true principal stresses. Therefore, to get the true 3-dimensional orientation of the principal stresses, at least three different orientations are required (such as testing three perpendicular planes or using a method like CSIRO). Once we get three crossing stresses, we can use the principal stresses rule to find a plane that has zero shear stress, and then get the perpendicular direction as the true orientation of the in-situ principal stresses.